

Foundations of the Geometric Vacuum Framework: A Synthesis of the Spectral Graph Theory Arc*

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The Geometric Vacuum (GeoVac) framework is a discretization scheme for non-relativistic quantum mechanics on natural geometries: the discrete graph Laplacian on a packed lattice of quantum-number labels replaces the continuous Laplace–Beltrami operator on the corresponding compact manifold. This synthesis paper assembles the framework’s foundational arc—nine papers covering the packing construction, the discrete graph and its dynamics, the conformal equivalence with the unit S^3 , the exchange constant taxonomy, the angular sparsity theorem, the Bargmann–Segal lattice, and the universal/Coulomb partition—into a single coherent narrative. Four headline structural results frame the program: (a) the discrete Fock graph is conformally equivalent to the unit S^3 Laplace–Beltrami operator, verified by 18 independent symbolic proofs; (b) the angular sparsity of two-body matrix elements depends only on $l_{\max}$ and is bit-identical across five distinct central potentials, qualifying as a potential-independent theorem about spherical-fermion bases; (c) the three-dimensional isotropic harmonic oscillator admits a discrete graph encoding on the Hardy space $H^2(S^5)$ that is bit-exactly π -free in exact rational arithmetic at every finite cutoff; (d) the framework’s results split cleanly along an A/D partition—an algebra-only universal sector that transfers free across central potentials, and a Dirac-operator-specific sector containing the spectrum, the calibration π , and the candidate fine-structure-constant combination of Paper 2. We present GeoVac as a discretization framework that exploits natural geometry, not as a new foundation for quantum mechanics: the discrete graph and the continuous Laplace–Beltrami operator are dual descriptions connected by proven conformal maps, and the interpretive question of which description is more fundamental is left open. The framework’s concrete computational advantages— $O(V)$ sparsity, integer eigenvalues on the unit S^3 , angular momentum selection rules baked into the basis, bit-exactly rational matrix elements on the Bargmann–Segal lattice, and the universal sparsity bound—are independent of that interpretive question.

I. INTRODUCTION

The standard formulation of non-relativistic quantum mechanics presupposes a continuous configuration space and treats the Schrödinger operator $-\frac{1}{2}\nabla^2 + V(r)$ as the fundamental object. For the hydrogen atom in particular, Fock showed in 1935 [1] that the momentum-space Schrödinger equation maps by stereographic projection onto the free Laplacian on the unit three-sphere S^3 . This observation has been developed extensively by Bargmann [2], Bander and Itzykson [3], Avery [8], Aquilanti and co-workers [9], and others working in the hyperspherical-momentum-space tradition. In each case, the S^3 structure is treated as an elegant reformulation of a known continuum problem: a useful change of variables for analytic work, not a foundation in its own right.

The Geometric Vacuum (GeoVac) framework [22, 23, 26] takes a complementary computational stance. Starting from a packing construction on an isotropic sphere, it builds a finite directed graph whose nodes are quantum-number labels (n, l, m) and whose edges are transition amplitudes generated by the $SU(2) \otimes SU(1,1)$ ladder algebra of hydrogen. The graph Laplacian on this lattice is dimensionless and scale-invariant; physical energies emerge only upon imposing the Fock energy-shell

constraint $p_0^2 = -2E$, which acts as a stereographic focal length. The framework’s central claim, established by Paper 7, is that this discrete construction converges to the unit S^3 Laplace–Beltrami operator in the continuum limit. The discrete and continuous descriptions are connected by a proven conformal equivalence, and we present them throughout as dual descriptions of the same physics; the interpretive question of ontological priority is explicitly left to the reader.

GeoVac differs from three nearby families of discretization schemes in ways worth naming at the outset. Lattice gauge theory discretizes the spacetime continuum on a hypercubic grid; GeoVac discretizes the quantum-number space and inherits the natural geometry of separable quantum systems. Tensor-network methods choose a basis adapted to entanglement structure; GeoVac chooses a basis adapted to angular momentum and the Fock conformal projection. Algebraic quantum field theory works at the level of operator algebras over Minkowski spacetime; GeoVac is non-relativistic and operates on compact Euclidean-signature manifolds. The rhetoric rule that governs the GeoVac papers (see the project’s positioning statement) applies here: GeoVac is not a new foundation for quantum mechanics, and the synthesis presented in this paper does not assert ontological priority of either the discrete or the continuum description.

* Group 3 synthesis in the Geometric Vacuum series

The Group 3 (foundations) arc comprises nine papers.

Paper 0 [22] states the packing axiom and derives the angular quantum numbers as packing indices. Paper 1 [23] sets up the spectral graph theory machinery and demonstrates exact reproduction of the Rydberg spectrum. Paper 6 [25] extends the framework to time evolution at $O(V)$ cost. Paper 7 [26] proves the conformal equivalence with the unit S^3 via 18 independent symbolic verifications and extends to the multi-electron S^{3N-1} setting. Paper 18 [27] classifies transcendental content in GeoVac observables via a six-tier exchange-constant taxonomy and identifies a master Mellin engine on the spectral side. Paper 21 [28] synthesizes the proof chain and frames the atomic spectra research program. Paper 22 [29] states the potential-independent angular sparsity theorem. Paper 24 [30] constructs the Bargmann–Segal lattice for the harmonic oscillator and establishes the four-layer Coulomb/HO asymmetry. Paper 31 [31] states the A/D universal/Coulomb partition in spectral-triple language.

Four headline structural results organize the rest of this synthesis. **(H1)** The discrete Fock graph is conformally equivalent to the unit S^3 at the level of the Laplace–Beltrami operator (Paper 7 [26]). **(H2)** The angular sparsity density of two-body matrix elements is potential-independent (Paper 22 [29]): a structural theorem of spherical-fermion bases, not a Coulomb-specific accident. **(H3)** The 3D harmonic oscillator admits a discrete graph encoding on the Hardy sector of S^5 that is bit-exactly π -free (Paper 24 [30]). **(H4)** The framework’s results decompose cleanly into a universal algebra-only sector and a potential-specific Dirac-dependent sector (Paper 31 [31]). We discuss the fine-structure constant combination $K = \pi(B + F - \Delta)$ of Paper 2 [24] only in the open-questions section, in keeping with the project rule that the combination rule remains conjectural at the rule level despite three independent spectral homes for B , F , and Δ .

II. THE PACKING CONSTRUCTION AND THE GRAPH

GeoVac’s discrete lattice is constructed by a packing argument that is prior to the Schrödinger equation [22]. Two axioms suffice.

Axiom 1 (binary distinguishability). A minimum of two distinguishable states is required to define a metric on an information-bearing manifold.

Axiom 2 (geometric indifference). In the absence of external constraints, the distribution of states on a homogeneous surface maximizes entropy and is therefore isotropic.

Applied to concentric isotropic shells in two dimensions, these axioms force the area-proportional state count on shell k to be $2k - 1$. The shell capacities $\{1, 3, 5, 7, \dots\}$ are isomorphic to the spherical harmonic basis Y_{lm} under the identification $l = k - 1$ and $m =$

$-l, \dots, +l$. The angular quantum numbers (l, m) and the per-shell degeneracy $2l + 1$ thus emerge from the packing geometry alone, without invoking the Schrödinger equation or the angular momentum operators $\hat{L}^2, \hat{L}_z$.

The principal quantum number n enters as a cumulative grouping of the first n shells, producing the count $\sum_{k=1}^n (2k - 1) = n^2$. Paper 0 is careful to flag that this grouping is natural but not uniquely forced by the two axioms. Similarly, the factor of two corresponding to spin multiplicity arises from the orientability of the compactified packing manifold (S^2), which the paper presents as a structural correspondence rather than a derivation. The strong claim is the angular sector (l, m) ; the other identifications are framed as structural correspondences.

From the packing to the graph

The packed lattice acquires a graph structure when the $SU(2) \otimes SU(1, 1)$ ladder operators of the hydrogen dynamical algebra [4] are imposed as edges:

$$L_{\pm}|n, l, m\rangle \propto |n, l, m \pm 1\rangle, \quad (1)$$

$$T_{\pm}|n, l, m\rangle \propto |n \pm 1, l, m\rangle. \quad (2)$$

The angular operators $L_{\pm}$ connect states within an n -shell along m ; the radial operators $T_{\pm}$ connect states between adjacent n -shells at fixed l . Paper 1 [23] demonstrates that the spectrum of the resulting graph Hamiltonian, constructed from the Casimir invariants of the algebra, reproduces the Rydberg formula $E_n = -1/(2n^2)$ exactly, with no fitting parameters. At the algebraic level the discrete and continuous descriptions therefore already agree on the spectrum.

The universal kinetic constant $\kappa = -1/16$

A second invariant of the packing geometry is the universal kinetic constant $\kappa = -1/16$. Its operational role is to map the dimensionless graph eigenvalues $\lambda_n = -(n^2 - 1)$ to the physical Rydberg spectrum via the energy-shell focal length $p_0 = Z/n$. Recent analysis [27] has clarified that $\kappa = -1/16$ is itself derivable from the Fock conformal projection rather than fitted: the squared coupling between adjacent n -shells in the Gegenbauer eigenbasis is $c^2(n, l) = (1/16)[1 - l(l + 1)/(n(n + 1))]$, giving $c^2(n, 0) = 1/16$ universally for $l = 0$. Equivalent readings identify $1/16$ as the squared Chebyshev transition amplitude, as the inverse Fock Jacobian $1/\Omega^4(0)$, and as the $l = 0$ base rate of the Casimir decomposition. Paper 18 reclassifies κ accordingly from *calibration* to *conformal* in the exchange-constant taxonomy.

Status of the packing-versus-physics question

The discrete lattice exists *a priori* from the packing construction, before any reference to the Coulomb po-

tential or the Schrödinger equation. Its persistent match to known atomic structure—the Rydberg spectrum at $n_{\max} \rightarrow \infty$, the chemical periodicity of multi-electron atoms (Paper 16, outside the foundations group), nuclear magic numbers under a harmonic-oscillator basis (Paper 23, applications group)—is therefore evidence that quantum-mechanical structure is consistent with a discrete underlying combinatorial topology, rather than a tautology. We present this as the *structural fit* between the packing construction and known atomic structure. The synthesis paper does not argue for the ontological priority of either description; the dual-description framing of Paper 7 governs the discussion.

III. THE S^3 CONFORMAL EQUIVALENCE

The theoretical foundation of GeoVac is the conformal equivalence between the discrete Fock graph and the unit S^3 Laplace–Beltrami operator established in Paper 7 [26]. This section states the result and explains its operational role.

The Fock projection and the chordal distance identity

Fock’s 1935 construction [1] writes the momentum-space hydrogen Schrödinger equation

$$\left[\frac{p^2}{2} - E \right] \psi(\mathbf{p}) - \frac{1}{2\pi^2} \int \frac{\psi(\mathbf{p}')}{|\mathbf{p} - \mathbf{p}'|^2} d^3p' = 0 \quad (3)$$

and applies the stereographic projection

$$u_i = \frac{2p_0 p_i}{p^2 + p_0^2}, \quad u_4 = \frac{p^2 - p_0^2}{p^2 + p_0^2}, \quad p_0 = \sqrt{-2E}, \quad (4)$$

which maps $\mathbf{p} \in \mathbb{R}^3$ onto the unit three-sphere $\mathbf{u} \in S^3 \subset \mathbb{R}^4$. The Coulomb kernel $1/|\mathbf{p} - \mathbf{p}'|^2$ becomes the inverse-square chordal distance on S^3 , and the integral equation reduces to the eigenvalue problem

$$-\Delta_{S^3} \Psi_n = (n^2 - 1) \Psi_n, \quad (5)$$

free-particle motion on the unit sphere. The Coulomb potential is absorbed into the curvature of the sphere; the spectrum is determined by the topology of S^3 alone.

The 18 symbolic proofs

Paper 7 [26] verifies the conformal-equivalence chain through 18 independent symbolic proofs implemented in *sympy*. The proofs cover the stereographic projection formulas, the unit sphere constraint, the chordal distance identity, the conformal factor identity, the conformal Laplacian transformation law, the integer eigenvalue spectrum, and the energy-shell projection from S^3 to $\mathbb{R}^3$.

These tests live in `tests/test_fock_projection.py` and `tests/test_fock_laplacian.py` in the GeoVac repository and are run as topological integrity checks before any release. Their status as machine-verified algebraic identities is a structural guarantee that the discrete-to-continuum bridge has no hidden steps.

Schrödinger recovery in the continuum limit

The graph Laplacian on the GeoVac lattice converges, in the continuum limit $n_{\max} \rightarrow \infty$ with appropriate rescaling, to the round S^3 Laplace–Beltrami operator. Imposing the energy-shell constraint $p_0 = Z/n$ then maps the unit-sphere eigenvalues $\lambda_n = -(n^2 - 1)$ to the hydrogenic Rydberg series $E_n = -Z^2/(2n^2)$. The dimensionless graph carries no scale; the Rydberg energy is injected exclusively at the projection step. This separation between dimensionless topology and dimensional projection is the *dimensionless vacuum* viewpoint of Paper 7: the calibration content of physical observables is concentrated entirely in the focal-length parameter p_0 .

Multi-electron extension to S^{3N-1}

Paper 7 [26] and Paper 21 [28] extend the construction to N -electron atoms by replacing S^3 with the unit sphere $S^{3N-1} \subset \mathbb{R}^{3N}$ under the $SO(3N)$ rotation group. The single-electron Fock projection becomes a multi-electron hyperspherical projection, and the N -fermion antisymmetry condition on S^{3N-1} reads as a representation-theoretic constraint on the angular sector. The chemical periodicity of the periodic table emerges from the S_N subgroup of $SO(3N)$ acting on the hyperspherical angular basis (see Paper 16, applications group, outside the foundations scope of this synthesis).

A particular consequence of the S^3 projection is the closed-form single-center two-electron repulsion integral $F^0(1s, 1s) = 5/8$, derived from the node-overlap formula for V_{ee} on S^3 in Paper 7. More generally, all single-center Slater F^k integrals reduce to exact rationals in arithmetic over $\mathbb{Q}$. The single-center two-electron problem is thus closed in rational arithmetic; transcendentals appear only at the projection-to-energy step (see Section VII).

What Paper 7 does and does not claim

Paper 7 is careful to separate established mathematics from new contribution. The stereographic projection formulas, the unit sphere constraint, the chordal distance identity, the conformal Laplacian relation, and the integer eigenvalue spectrum are all known results from Fock, Bargmann, Bander–Itzykson, and Barut–Kleinert. The new contributions are: (a) the explicit convergence of the discrete GeoVac graph to the continuous S^3 Laplace–Beltrami operator; (b) the systematic symbolic verifi-

TABLE I. Angular ERI density $D(l_{\max})$ from Paper 22: fraction of orbital quartets (a, b, c, d) for which the Gaunt coefficient does not vanish identically. Universal across Coulomb, harmonic oscillator, Woods–Saxon, square well, and Yukawa potentials.

$l_{\max}$	$D(l_{\max})$
0	100%
1	7.81%
2	2.76%
3	1.44%
4	0.90%
5	0.62%

cation of the conformal transformation chain; (c) the framing of physical scales as confined to the energy-shell projection; (d) the computational consequence that $O(V)$ sparse graph eigenvalue problems can replace continuous integration. Paper 7’s explicit statement of the rhetoric—“the mathematics supports both the discrete-as-fundamental and the continuous-as-fundamental readings, and we leave the choice to the reader”—is the foundation of the dual-description framing carried through the rest of the foundations papers.

IV. ANGULAR SPARSITY AS A STRUCTURAL THEOREM

The angular sparsity theorem of Paper 22 [29] is the foundations-side statement of the structural mechanism behind GeoVac’s $O(Q^{2.5})$ Pauli-term scaling at the qubit-encoding level. The theorem can be stated cleanly without reference to the Coulomb potential or to any specific radial wavefunction.

Theorem (Paper 22). *For any N -fermion system whose single-particle orbitals factor as products of spherical harmonics and arbitrary radial functions, the fraction of nonzero two-body matrix elements is determined solely by angular momentum selection rules and is independent of the radial potential.*

The proof is a direct application of the Wigner–Eckart theorem and Condon–Shortley angular momentum algebra [14, 15]. The two-body matrix element of a rotationally invariant interaction factors into a sum of products of Gaunt coefficients (angular factor) and Slater F^k, G^k integrals (radial factor); the Gaunt coefficients vanish for quartets (l_a, l_b, l_c, l_d) that do not satisfy the $3j$ -triangle inequality $|l_a - l_c| \leq k \leq l_a + l_c$ and the corresponding inequality for (l_b, l_d) . Exact enumeration at fixed $l_{\max}$ yields the ERI densities reproduced in Table I.

Universality across potentials

The bound has been verified numerically across five qualitatively distinct potentials—Coulomb $-Z/r$,

isotropic harmonic oscillator $\frac{1}{2}m\omega^2 r^2$, Woods–Saxon nuclear single-particle potential, square well, and Yukawa $e^{-\mu r}/r$ —and produces bit-identical sparsity patterns at matched orbital counts. The universality is structural: the radial-angular factorization of $1/r_{12}$ in spherical harmonics (used in quantum chemistry) and the Talmi–Moshinsky bracket factorization of central NN potentials in oscillator bases (used in nuclear shell models) impose the same angular selection rules. GeoVac’s contribution is to state the explicit quantitative density at each $l_{\max}$ and to frame it as a universal resource guarantee independent of the radial physics.

Spinor extension

Paper 22’s spinor extension shows that the angular sparsity bound survives, with mildly inflated prefactor, when the basis is lifted from Y_{lm} to the spinor spherical harmonics $\Omega_{\kappa m_j}$ labeled by the Dirac quantum numbers (κ, m_j) . The density $d_{\text{spinor}}(l_{\max})$ satisfies $d_{\text{spinor}} \leq d_{\text{scalar}}$ at every $l_{\max}$ tested, with the same triangle inequality controlling vanishing of the matrix elements at the spinor level. Tier-2 spin-orbit extensions of the framework therefore inherit the angular sparsity guarantee without modification.

Operational consequence

The angular sparsity theorem is what makes GeoVac’s qubit-encoded Hamiltonians competitive for VQE/NISQ resource estimation. The quantum-computing applications papers (Group 4) report Pauli-term counts that scale as $O(Q^{2.5})$ for the composed-geometry architecture across LiH, BeH₂, and H₂O—a $51\times$ to $1712\times$ advantage over published Gaussian baselines at matched qubit count. The structural reason for this advantage is the Gaunt-selection-rule sparsity quantified in Table I; the foundations-paper statement is the theorem itself, not the specific molecular benchmarks.

V. QUANTUM DYNAMICS ON THE GRAPH

Paper 6 [25] extends the framework from spectroscopy to time evolution. The graph Hamiltonian H is sparse with $O(V)$ nonzero entries, so the Crank–Nicolson propagator

$$\psi(t + \Delta t) = (I + \frac{i}{2}H\Delta t)^{-1} (I - \frac{i}{2}H\Delta t) \psi(t) \quad (6)$$

admits implementation at $O(V)$ cost per step using sparse matrix-vector products plus iterative linear solves. Unitarity is preserved to machine precision over 10^4 time steps in the H₂ benchmark reported by Paper 6.

The applications demonstrated in Paper 6 are: (a) broadband molecular spectroscopy from a single delta-kick propagation, recovering 20 dipole-active electronic

transitions of H_2 with 0.16% mean error in 33 seconds; (b) resonant Rabi oscillations matching the analytic Bloch–Siegert-corrected period to 0.41% with 99.98% coherent population transfer; (c) ab initio molecular dynamics with nuclei coupled to graph forces via Velocity Verlet, conserving total energy to 0.0003%; (d) NVT-ensemble Langevin dynamics with stable thermal vibrations at 300 K.

Paper 6 is not a foundational claim in the same sense as Papers 7, 18, 22, 24, or 31—it is an operational demonstration that the foundations support time evolution. Its role in the synthesis is to close the gap between the spectral statement of Papers 1 and 7 (the graph reproduces the Rydberg spectrum) and the dynamical statement required for any quantum-chemistry application (the graph supports time evolution at the same $O(V)$ cost). Without Paper 6’s dynamics engine, GeoVac would be limited to eigenvalue problems; with it, the framework extends to time-dependent quantum chemistry, spectroscopy, and molecular dynamics.

VI. THE BARGMANN–SEGAL LATTICE AND THE COULOMB/HO ASYMMETRY

Paper 24 [30] extends GeoVac to the three-dimensional isotropic harmonic oscillator (HO) and uncovers a structural asymmetry between the Coulomb and HO discretizations. The construction is built on the Bargmann–Segal transform [5, 6].

The discrete HO lattice on S^5

The Bargmann–Segal transform maps the 3D isotropic HO Hamiltonian to the Euler number operator $\hat{N}$ on the Fock–Bargmann space of holomorphic functions on $\mathbb{C}^3$. Restriction to the unit sphere $S^5 \subset \mathbb{C}^3$ identifies the eigenspaces with the symmetric $SU(3)$ irreducible representations $(N, 0)$, of dimension $(N+1)(N+2)/2$, which match the HO shell degeneracies exactly. The HO Hamiltonian becomes $\hbar\omega(\hat{N}+3/2)$, with the eigenvalues linear in the shell index N .

The discrete graph encoding of this structure takes nodes (N, l, m_l) to be the labels of the $(N, 0)$ -restricted hyperspherical harmonics on S^5 , and edges to be dipole transitions $\Delta N = \pm 1$, $\Delta l = \pm 1$ at exact-rational squared matrix elements. At $N_{\max} = 5$ the graph has 56 nodes and 165 edges, all matrix elements computed in exact `Fraction` arithmetic, with zero irrationals encountered. The single π in the entire Bargmann–Segal ecosystem is the Gaussian normalization π^{-3} of the continuous Bargmann measure, which the discrete graph never invokes.

The HO rigidity theorem

Theorem (Paper 24, HO rigidity). *The 3D isotropic harmonic oscillator is the unique central potential whose spectrum arises from the Euler number operator on the Hardy space $H^2(S^5)$ restricted to symmetric $SU(3)$ irreps $(N, 0)$.*

This is the structural dual of the Fock rigidity theorem of Paper 23 (applications group), which states that the S^3 stereographic projection $p_0 = \sqrt{-2E_n}$ maps a one-electron central-field Hamiltonian onto the free Laplacian on S^3 if and only if the spectrum is l -independent within each n -shell, a property unique to $-Z/r$ by $SO(4)$ symmetry. Together the two rigidity theorems establish that the choice of central potential is sharply tied to the choice of natural manifold: $-Z/r$ to the second-order Riemannian Laplacian on S^3 , $\frac{1}{2}m\omega^2 r^2$ to the first-order Euler operator on $H^2(S^5)$.

The four-layer Coulomb/HO asymmetry

A central structural finding of Paper 24, sharpened by subsequent analysis in Paper 25 (synthesis group) and a recent sprint on the wedge KMS state of the framework’s Lorentzian extension, is that the Coulomb and HO discretizations differ along four independent layers. The asymmetry as currently catalogued reads:

1. **Spectrum-computing role of L_0 .** On the Coulomb S^3 graph, the graph Laplacian L_0 is the spectrum-generating operator: $(D - A)$ produces the hydrogenic spectrum $n^2 - 1$. On the Bargmann–Segal lattice, L_0 is the matter-propagator operator, not the Hamiltonian: the spectrum comes from the diagonal Euler operator $\hbar\omega(\hat{N}+3/2)$, and the graph adjacency encodes dipole transitions only.
2. **Calibration π .** The Coulomb S^3 projection introduces calibration π as the Hopf-base measure (see Section VII); the HO Bargmann projection is bit-exactly π -free at every finite $N_{\max}$. This is one of the clearest concrete instances of an exchange constant being tied to a specific natural geometry.
3. **Wilson-physical-content.** The non-abelian Wilson gauge construction on $S^3 = SU(2)$ admits natural matter coupling (the gauge field acts on Pauli spinors carried by the graph nodes); the parallel $SU(3)$ Wilson construction on the $(N, 0)$ Hardy tower fails the fixed-group-on-every-link requirement, because transitions between $(N, 0)$ and $(N+1, 0)$ irreps are Clebsch–Gordan intertwiners rather than $SU(3)$ group elements. The natural gauge group of the Bargmann graph at the pure-gauge level is $U(1)$, not $SU(3)$.

4. **Modular-Hamiltonian structure of the wedge KMS state.** The Coulomb S^3 side admits a hemispheric-wedge KMS state with closed-form Hilbert–Schmidt orthogonality between the modular Hamiltonian and the wedge-restricted Dirac operator, with prefactor $1/\pi^2$ identifying as the Hopf-base measure signature; the HO Bargmann side does not admit the construction at all. This is the operator-algebra-side reading of the four-layer asymmetry; for full details see the operator-algebra arc (Group 1).

The first three layers were established in Paper 24 and Paper 25 (synthesis group, S^5 gauge structure paper); the fourth layer was identified in a 2026 sprint on the wedge KMS state and folded into Paper 24’s §V as an addendum. All four layers stratify along the same operator-order divide: the Coulomb side uses a second-order Riemannian Laplacian with a nonlinear stereographic projection, and the HO side uses a first-order complex-analytic Euler operator with a linear-affine Bargmann projection.

What the asymmetry says about transcendentals

The bit-exact π -freeness of the Bargmann–Segal lattice is a structurally clean statement about where transcendentals enter physical observables. π is not a universal feature of natural geometry discretizations; it is a specific structural cost of the S^3 stereographic projection. This refines the exchange-constant taxonomy of Paper 18 (see Section VII): calibration π is Coulomb-specific, tied to second-order Riemannian operators with nonlinear projections, and has no analog on the first-order HO side.

VII. THE EXCHANGE CONSTANTS TAXONOMY

Paper 18 [27] catalogues where transcendental constants enter physical observables computed on the GeoVac graph. The taxonomy is a six-tier classification organized by the type of compactness mismatch each transcendental mediates between the discrete representation and the non-compact coordinate system used for measurement.

Six tiers

1. **Intrinsic.** Algebraic content native to the discrete graph: rational eigenvalues, integer multiplicities, exact-rational Slater F^k integrals on S^3 . π -free in arithmetic over $\mathbb{Q}$.
2. **Conformal / calibration.** π entering through the Hopf measure of the stereographic projection; the

universal kinetic constant $\kappa = -1/16$; the Hopf-base measure $\text{Vol}(S^2)/4 = \pi$ identifying as the a_0^2 Seeley–DeWitt coefficient on the spinor bundle. This is the source of calibration π in the Coulomb sector.

3. **Embedding.** Transcendentals introduced by embedding the discrete graph into a continuum coordinate system that the graph does not natively know: the electron-electron cusp $1/r_{12}$ is the canonical example. The cusp is two-dimensional in the angular pair coordinate (α, θ_{12}) and cannot be reproduced by any single algebraic insertion; partial-wave Schwartz extrapolation $\sim -A/(l_{\max} + 2)^4$ is the standard treatment.
4. **Algebraic-implicit.** Quantities defined by a polynomial equation $P(R, \mu) = 0$ with coefficients in a known field extension, where pointwise diagonalization is a computational convenience rather than a mathematical necessity. The Level-3 adiabatic potential curves $\mu(R)$ on the hyperspherical fiber bundle live in $\mathbb{Q}(\pi, \sqrt{2})[R]$.
5. **Composition.** Single transcendental seeds appearing through algebraic reduction of integrals: the seed $e^a \cdot E_1(a)$ that closes the Level-2 prolate-spheroidal radial solver via partial fraction decomposition; the seed $\gamma(1, x)$ closing the Slater F^k Laguerre-moment decomposition.
6. **Inner-factor input data.** Parameter-tied transcendental content that the framework consumes as Layer-2 input rather than deriving from Layer-1 graph data. The Yukawa Dirichlet ring $\mathbb{Q}[y_1^{-2s}, \dots, y_n^{-2s}]$ associated with the inner-factor of a Connes–Marcolli almost-commutative spectral triple is the canonical example; nuclear-structure scalars $(\langle r^2 \rangle_E, \text{Zemach radius } r_Z, \text{quadrupole moment } Q_N)$ are also parameter-tied Layer-2 inputs. This tier captures the boundary between what GeoVac derives and what it consumes from external spectroscopy.

The taxonomy is falsifiable: the claim is that the transcendental content of any GeoVac observable is determined entirely by the type of projection linking the compact discrete description to the non-compact coordinate system. No observable has been identified that violates the taxonomy; recent work on bound-state QED in the hydrogen-Lamb-shift sprint has confirmed that the framework-native one-loop contributions sit cleanly in the calibration and composition tiers, with multi-loop UV-renormalization counterterms moving the relevant content into the inner-factor input data tier.

The master Mellin engine

A structural sharpening of the taxonomy is the master Mellin engine identified in Paper 18 §III.7 and formalized in the operator-algebra arc of the project. The Weyl–Selberg identification of GeoVac’s exchange constants with Mellin transforms of heat kernels

$$\mathcal{M}\left[\mathrm{Tr}\left(D^k \cdot e^{-tD^2}\right)\right](s) \quad (7)$$

indexes three sub-mechanisms by the integer operator order $k \in \{0, 1, 2\}$:

- **M1** ($k = 0$). Trivial heat kernel collapses to the Hopf-base measure $\mathrm{Vol}(S^2)/\pi^2 = 4/\pi$. This is the signature of all calibration- π content in the Coulomb sector and appears as the leading constant in the GeoVac propinquity convergence rate.
- **M2** ($k = 2$). Standard Seeley–DeWitt expansion of $\mathrm{Tr} e^{-tD^2}$. Produces $\sqrt{\pi} \mathbb{Q} \oplus \pi^2 \mathbb{Q}$ at every integer s via Jacobi θ -inversion on the half-integer-shifted Dirac spectrum.
- **M3** ($k = 1$). Mellin transform of $\mathrm{Tr}(D \cdot e^{-tD^2})$. Produces the quarter-integer Hurwitz / vertex-parity content (Catalan’s G , Dirichlet $\beta(2k)$) that appears in two-loop QED vertex corrections on S^3 .

The master Mellin engine reading is that M1, M2, M3 are not three independent mechanisms but three sub-cases of the same Mellin transform on the GeoVac spectral triple, indexed by which power of the Dirac operator appears in the trace. Each sub-mechanism has a characteristic domain of observable: M1 governs propinquity convergence rates and Hopf-measure normalizations, M2 governs heat-kernel and spectral-action coefficients, and M3 governs vertex-restricted parity-character observables. Mixing the operator order of the question with the operator order of the answer produces the structural degeneracies catalogued in the project’s sprint record.

The Layer-1 / Layer-2 split

A particularly clean operational consequence of the taxonomy is the two-layer architecture made explicit by Paper 34 (catalogue paper, applications group): the bare combinatorial graph (Layer 1, π -free in $\mathbb{Q}[\sqrt{d}]$) versus a finite family of named projections (Layer 2) that add specific variables, dimensions, and transcendentals when the graph is mapped to a physical observable. Paper 34 documents 28 projections to date; each appears in the taxonomy of Paper 18 as either an intrinsic, calibration, embedding, algebraic-implicit, composition, or inner-factor input data mechanism. The two papers are complementary: Paper 18 organizes by type of compactness mismatch (what *kind* of transcendental appears), and Pa-

per 34 organizes by named mechanism (*which* projection appears in *which* observable).

VIII. THE UNIVERSAL/COULOMB PARTITION

Paper 31 [31] states the cleanest single articulation of what GeoVac claims as universal versus what it claims as potential-tied. The framework is read as a discrete spectral triple $(\mathcal{A}, \mathcal{H}, D)$ in the sense of Connes [10] and the graph-spectral-triple lineage of Marcolli and van Suijlekom [11, 12]: $\mathcal{A}$ is an algebra of functions on the Fock-projected discrete graph, $\mathcal{H}$ is a Hilbert space of scalar and spinor states, and D is a Dirac operator inherited from the Camporesi–Higuchi spectrum on S^3 [7].

The A/D split

The structural decomposition reads:

Universal sector ($\mathcal{A}$ -only). Results that depend only on the algebra $\mathcal{A}$ and the angular structure of the basis, and *not* on the specific choice of Dirac operator. These transfer free across central potentials. Examples:

- Angular sparsity (Paper 22): ERI density $D(l_{\max})$ depends only on the spherical-harmonic angular factorization.
- Composed scaling (Paper 14, applications group): the $O(Q^{2.5})$ Pauli-term scaling of composed-geometry qubit encodings depends on the Gaunt selection rules.
- Selection rules: $\Delta l = \pm 1$, $\Delta m = 0, \pm 1$ for dipole transitions; the $3j$ -triangle inequality at every vertex of the two-body integral.

Potential-specific sector (D -dependent). Results that depend on the specific spectrum and matrix elements of the Dirac operator D , and therefore on the specific central potential. These require per-system rederivation. Examples:

- The spectrum itself: $n^2 - 1$ for Coulomb, $N + 3/2$ for HO.
- The calibration $\kappa = -1/16$ (Coulomb-specific; no analog on the Bargmann lattice).
- Transcendental content: π for Coulomb via the Hopf measure, π -free for HO.
- Rigidity theorems: Fock rigidity (Paper 23) pins the Coulomb geometry to S^3 ; HO rigidity (Paper 24) pins the HO geometry to $H^2(S^5)$ with $(N, 0)$ symmetric irreps.

- The candidate fine-structure-constant combination $K = \pi(B + F - \Delta)$ of Paper 2: lives entirely in the Coulomb-specific sector by its dependence on the Camporesi–Higuchi Dirac spectrum on S^3 .

Why the partition matters operationally

The partition organizes Papers 22, 23, 24, and 25 into a single conceptual structure. Paper 22’s angular sparsity theorem lives in the universal sector and transfers free between hydrogen and the deuteron; Paper 23’s nuclear shell model qubit Hamiltonians inherit the angular sparsity automatically but require a fresh per-potential analysis of the Dirac sector. Paper 24’s HO rigidity and the π -free certificate live in the potential-specific sector and explicitly do *not* transfer to the Coulomb side—the Bargmann-style first-order operator has no analog on S^3 . Paper 25 (synthesis group) on the $SU(2)$ Wilson lattice gauge structure of the Hopf graph also lives in the potential-specific sector: it depends on the parallelizability of $S^3 = SU(2)$, which has no analog on S^5 .

The operational utility of the partition is that anyone porting GeoVac to a new system can read off, from the universal sector, what structural properties they inherit for free, and from the potential-specific sector, what calibrations they need to derive afresh. For nuclear shell models, gauge-theoretic extensions, and oscillator-based quantum dots, the universal sector provides angular sparsity and composed scaling automatically; the potential-specific sector has to be redone for each system.

Status as a spectral-triple-language reformulation

Paper 31 is careful to frame the partition as a structural reformulation, not a new theorem. The angular sparsity theorem (Paper 22), the Fock rigidity theorem (Paper 23), and the HO rigidity theorem (Paper 24) are each established results; the partition is the observation that they decompose cleanly along the algebra-only / Dirac-dependent divide. The framing uses the language of NCG and spectral triples because that language has the structural vocabulary to articulate the divide; the underlying physics content is unchanged.

IX. OPEN QUESTIONS AND OUTLOOK

The foundations arc as catalogued here leaves four named structural questions open.

The α combination rule

Paper 2 [24] (observations group) presents the candidate combination

$$K = \pi(B + F - \Delta), \quad K \stackrel{?}{=} \alpha^{-1}, \quad (8)$$

where $B = 42$ is a finite Casimir trace at the truncation $m = 3$, $F = \zeta_R(2) = \pi^2/6$ is the Fock-degeneracy Dirichlet series at the packing exponent $d_{\max} = 4$, and $\Delta = 1/40$ is the inverse Dirac mode count $g_3^{\text{Dirac}}(S^3) = 2(n+1)(n+2)|_{n=3}$. Each ingredient now has a structurally identified spectral home; twelve mechanisms have been eliminated as candidate single-principle derivations of the sum [27]. The combination rule itself remains conjectural at the rule level: there is no first-principles derivation explaining why the additive combination of three structurally independent objects equals α^{-1} at 8.8×10^{-8} . The synthesis treats this as an open structural question and does not assert priority for the conjectural identification.

The second packing axiom (W3)

The packing construction of Paper 0 generates the angular sector (l, m) and provides natural correspondences for n and spin multiplicity, but it does not produce calibration data such as Yukawa couplings or the number of fermion generations. These appear in the exchange-constant taxonomy as Layer-2 inner-factor input data: the framework consumes them rather than deriving them. The open question labeled W3 in the project’s internal taxonomy is whether a second packing-axiom-like construction generates calibration data from a similarly minimal principle. A 2026 sprint surveying candidates in the spectral-zeta / master-Mellin-engine ring returned uniformly negative results across PMNS angles, lepton masses, and a $\sim 21,000$ -form mechanical basis [32]. The second-packing-axiom question is open and is structurally distinct from “find a packing on a different geometry”—the latter has been answered for the HO via the Bargmann–Segal lattice without addressing W3.

Cross-manifold extensions

The Coulomb and HO discretizations live on S^3 and the Hardy sector of S^5 respectively. A natural question is whether a coupled system (electronic plus nuclear, or two distinct potential regimes within one molecule) admits a tensor-product spectral triple of the form $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\text{Hardy}(S^5)}$. Within the foundations arc, Paper 24’s four-layer Coulomb/HO asymmetry documents structural obstructions to such a construction at the NCG-framework level. The operator-algebra arc (Group 1) has formally

named this as a frontier-of-field open problem; no published framework currently supports the cross-manifold tensor product.

The multi-focal composition wall

The framework’s composition rules for multiple Fock-style projections constitute an active research direction. The internal taxonomy labels three sub-walls: W1 (within-framework architectural gaps: cross-register coordinate operators, magnetization-distribution operators, frozen-core cross-center screening), W2 (multi-loop UV/IR composition and cross-manifold extensions), and W3 (inner-factor calibration data, discussed above). W1a/b/c have been closed at the first-pass level by the Phase C work documented in the applications group; W2a (multi-loop renormalization) remains open at the field-theory frontier of the operator-algebra arc. None of these walls falsify the foundations as presented in this synthesis; they sharpen the open-question boundary of the framework.

X. CONCLUSION

The Group 3 foundations arc of GeoVac establishes a small set of structural results that together provide a coherent discretization of non-relativistic quantum mechanics on natural geometries. The discrete Fock graph is provably conformally equivalent to the unit S^3 Laplace–Beltrami operator (Paper 7 [26]); the angular sparsity of two-body matrix elements is potential-independent (Paper 22 [29]); the 3D harmonic oscillator admits a bit-exactly π -free discrete encoding on $H^2(S^5)$ (Paper 24 [30]); the framework’s results decompose cleanly into an algebra-only universal sector and a Dirac-dependent potential-specific sector (Paper 31 [31]); the exchange-constant taxonomy of Paper 18 [27] catalogues where transcendentals enter physical observables and identifies a master Mellin engine at three operator

orders as the generator of all known π -content; the packing axiom of Paper 0 [22] generates the angular sector geometrically; the spectral graph theory of Paper 1 [23] and the $O(V)$ time evolution of Paper 6 [25] provide the computational machinery.

The framework’s concrete advantages are independent of the interpretive question of which description—discrete or continuous—is more fundamental. $O(V)$ sparsity, integer eigenvalues on the unit S^3 , angular momentum selection rules baked into the basis, bit-exactly rational matrix elements on the Bargmann–Segal lattice, and the universal sparsity bound at fixed $l_{\max}$ are statements about the discrete graph; their continuum counterparts are theorems about the Laplace–Beltrami operator or the Euler operator on the corresponding natural manifold; the conformal equivalence guarantees consistency. The discrete and continuous readings are dual descriptions connected by proven maps, and the interpretive question is left open.

The open questions catalogued in Section IX are substantive but do not destabilize the structural results of the foundations arc. The α combination rule remains conjectural; the second-packing-axiom question is open; cross-manifold extensions are blocked at the NCG-framework level; the multi-focal composition wall is an active research frontier. Each of these is a frontier-of-field question rather than a foundational concern, and the synthesis treats them as such.

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